

Global distribution and genomic characteristics analysis of avian-derived *mcr-1*-positive *Escherichia coli*

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ABSTRACT

The prevalence of avian-derived *Escherichia coli* (*E. coli*) carrying *mcr-1* poses a significant threat to the development of the poultry industry and public health safety. Despite ongoing in-depth epidemiological research worldwide, a comprehensive macroscopic study based on genomics is still lacking. In response, this study collected 1104 genomic sequences of avian-derived *mcr-1*-positive *E. coli* (MCRPEC) from the NCBI public database, covering 31 countries. The majority of sequences originated from China (48.82 %), followed by the Netherlands (10.41 %). In terms of avian hosts, chicken accounted for the largest proportion (44.11 %), followed by gallus (24.09 %). Avian-derived MCRPEC also serves as a reservoir for other antibiotic resistance genes (ARGs), with 179 ARGs coexisting with *mcr-1* identified. A total of 206 virulence-associated genes were also identified, revealing the pathogenic risks of MCRPEC. Pan-genome analysis revealed that avian-derived MCRPEC from different hosts, countries of origin, and serotypes exhibit minor SNP differences, indicating a high risk of cross-regional and cross-host transmission. The ST types of MCRPEC are diverse, with ST10 being the most prevalent (n=70). Spearman analysis showed a significant correlation between the number of ARGs and the insertion sequences (ISs) as well as plasmid replicon in ST10 strains. Furthermore, ST10 strains share a similar genetic basis with human-derived MCRPEC, suggesting the possibility of clonal dissemination. Pan-genome-wide association studies (pan-GWAS) indicated that the differential genes of MCRPEC from different countries and host sources are significantly different, mainly related to genes encoding type IV secretion systems and mobile genetic elements (MGEs). Plasmid mapping of showed that the prevalent plasmid types vary by country and host, with Inc12 and IncX4 being the main *mcr-1*-positive plasmids. Among the 12 identified *mcr-1* genetic contexts with ISs, the Tn6330 transposon was the predominant carrier of *mcr-1*. In summary, the potential threat of avian-derived MCRPEC cannot be ignored, and long-term and comprehensive monitoring are essential.

1. Introduction

The transmissions of ARGs poses a significant risk to public health. The last bastion against multidrug-resistant bacteria includes the polymyxin, carbapenems, and novel tetracyclines. However, the rapid

spread of plasmid-mediated ARGs has severely compromised this final line of defense, with these genes now being recognized as emerging environmental contaminants. Since the landmark report of the plasmid-mediated polymyxin resistance gene *mcr-1* in China in 2015, subsequent research has uncovered its presence across various regions and hosts,

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underscoring the substantial threat to the management of antibiotic resistance (Shen et al., 2018; Liu et al., 2016; Zhang et al., 2021). In response to this escalating challenge, China planned to prohibit the use of polymyxins in animal feed and health products by 2017. (Walsh and Wu, 2016). The previous widespread use of polymyxin-enriched feed in animal agriculture was identified as a principal driver of the *mcr-1* dissemination. Encouragingly, research indicates that the implementation of this ban has coincided with a marked decrease in *mcr-1* positive samples across diverse regions and sources in China (Shen et al., 2020). Although the trade of colistin has been prohibited and its usage restricted in several countries, the persistence of colistin use poses a significant challenge to the containment of *mcr-1* spread worldwide (Umair et al., 2023). It is essential to enforce sustained control measures and monitoring to mitigate this issue.

E. coli is widely present and transmitted in humans, animals, and the environment. Due to its MGE promoting the rapid and widespread spread of ARGs, it is a key target for monitoring these genes (Zhang et al., 2019, 2021, 2023). Poultry, constituting a significant segment of global agricultural trade, has reportedly experienced a notable upswing in trade activity post the COVID-19 pandemic in 2019, as indicated by the United States Department of Agriculture data (<https://fas.usda.gov/data/livestock-and-poultry-world-markets-and-trade>, available in September 2024). This intense and voluminous trade activity is also a critical driver for the swift dissemination of ARGs such as *mcr-1*. Research indicates that poultry serve as the principal reservoir for *mcr-1*-positive *Enterobacteriaceae*, with identified risks of *mcr-1* transmission between poultry and humans (Kluytmans-van den Bergh et al., 2016; Trung et al., 2017; Zhang et al., 2020).

Research into bacterial antibiotic resistance is underpinned by the swift advancement of whole-genome sequencing technologies, complemented by bioinformatics analytical tools. These contemporary methodologies have not only heightened precision compared to prior molecular detection techniques and phylogenetic analyses but also facilitated deeper exploration into the genomics of drug-resistant bacteria, leading to a more holistic elucidation of their epidemiological dissemination (Shen et al., 2019). Second-generation bacterial draft sequencing technology, while more cost-effective, yields a more comprehensive dataset for bacterial strains, making it indispensable for large-scale genomic analyses. The evolution of bioinformatics techniques has further deepened and broadened the excavation of bacterial genomic data. Current literature indicates that genomic-based analyses of antibiotic-resistant bacteria can reveal their epidemiological traits, providing valuable perspectives for future research and preventative strategies (Cai et al., 2024; Yu et al., 2024).

MCRPEC has garnered extensive global attention, with genomics-based research revealing a heterogeneous genetic basis and intricate geographic spread, leading to diverse prevalence across nations, thereby increasing the challenge of *mcr-1* containment (Li et al., 2022; Pan et al., 2022; Li et al., 2020). Present epidemiological studies on *mcr-1* are predominantly centered on human clinical contexts and other animal reservoirs, while the majority of epidemiological investigations into avian-derived *mcr-1* are subject to regional constraints and limitations in multi-dimensional analysis. Our study leverages the genomic data on avian-derived MCRPEC available in the NCBI database, applying a suite of bioinformatics analytical tools to deeply explore and scrutinize the genomic datasets. The objective is to establish a theoretical framework for the distribution and dissemination of avian-derived MCRPEC and to provide referential value for the strategic management of *mcr-1* transmission within avian populations.

2. Methods

2.1. Screening of avian-derived *mcr-1*-positive *E. coli* in public databases

As of February 28, 2024, a quantitative analysis of avian-derived *E. coli* was conducted in the NCBI Pathogen Detection database

(<https://www.ncbi.nlm.nih.gov/pathogens>). The screening criteria included "AMR genotypes core: *mcr-1*" and "TaxID: 562", with a focus on selecting host sources related to avian species. All collected sequences were quality-checked using CheckM v1.2.2 (Parks et al., 2015), retaining only high-quality genomic data. A total of 1104 avian-derived MCRPEC strains were collected, with basic information such as strain serial numbers, country of origin, host source, and isolates collection time documented.

2.2. Bioinformatics analysis

The MLST v.2.23.0 (<https://github.com/tseemann/mlst>) was utilized to perform Multilocus Sequence Typing for all strains. The local version of PHYLOViZ v.2.0 was applied to construct and visualize the phylogenetic tree of the obtained ST types (Nascimento et al., 2017). Resfinder v.4.5.0 was employed with default parameters for annotating resistance genes (Bortolaia et al., 2020). ECTyper v.1.0.0 was used to distinguish serotypes (Bessonov et al., 2021). ClermonTyping was applied for systematic typing of the strains (Beghain et al., 2018). The ABRicate tool with VFDB was selected for annotating VFs in the strains (Chen et al., 2016). Plasclass v.0.1.1 was used for *mcr-1* localization, with a threshold >0.8 indicating the presence on plasmids (Pellow et al., 2020). ISEScan v.1.7.2.3 was used for the identification and annotation of ISSs (Xie et al., 2017). PlasmidFinder v.2.1.6 was employed for the annotation of plasmid replicon types (Carattoli and Hasman, 2020). Prokka v.1.14.6 was run with autoprotokka (<https://github.com/stevenjduffy/autoprotokka>) for batch annotation (Seemann, 2014). The pan-genome analysis was conducted using Roary v.3.13.0, yielding core genome alignment files and files indicating the presence and absence of genomes (Page et al., 2015). Snp-dists v.0.8.2 was used to analyze SNP differences in the core genome between each pair of strains (<https://github.com/tseemann/snp-dists>). Based on the SNP differences in the core genome, a phylogenetic tree was constructed using FastTree v.2.1.11 and visualized and beautified with Evolview (Price et al., 2009; He et al., 2016). Scoary was used for pan-GWAS, with Bonferroni p set to less than or equal to 0.01 to prevent false positives (Brynildsrud et al., 2016). Cdhit v.4.8.1 was used for sequence de-replication. Easyfig v.2.2.5 was employed for creating linear comparison graphs of genes (Sullivan et al., 2011). Mob-suite v.3.1.9 was utilized for plasmid typing of the sequences (Robertson and Nash, 2018).

2.3. Statistical analysis

The results of the aforementioned analyses were extracted and merged using Python. Box plots were created using the online website Chiot, with t-tests for significance analysis, principal component analysis using the Bray-Curtis algorithm, and confidence circles constructed based on a 95 % confidence level (<https://www.chiotplot.online/>). The Veen network co-occurrence plot was used on the online website Veen for co-existence analysis of three countries and three types of hosts (Chen et al., 2021). The ggstatsplot package in R v.4.3.3 was utilized for Spearman analysis and visualization of ARGs, ISSs, VFs, and plasmid replicons. Gephi v.0.10.1 was employed for the generation of co-occurrence networks of ARGs (<https://gephi.org/>).

3. Result

3.1. Sources of avian-derived MCRPEC

A total of 1104 avian-derived MCRPEC strains were collected from NCBI (Table S1). The earliest collection of avian-derived MCRPEC was in 1975, isolated from the cloaca of chickens in China (Wu et al., 2018). Since 2010, there has been a sharp increase in the number of MCRPEC sequences in the NCBI database, with the highest number of positive *mcr-1* strains collected and uploaded in 2020. In this study, MCRPEC were isolated from nine different described avian sources (Fig. 1C), with

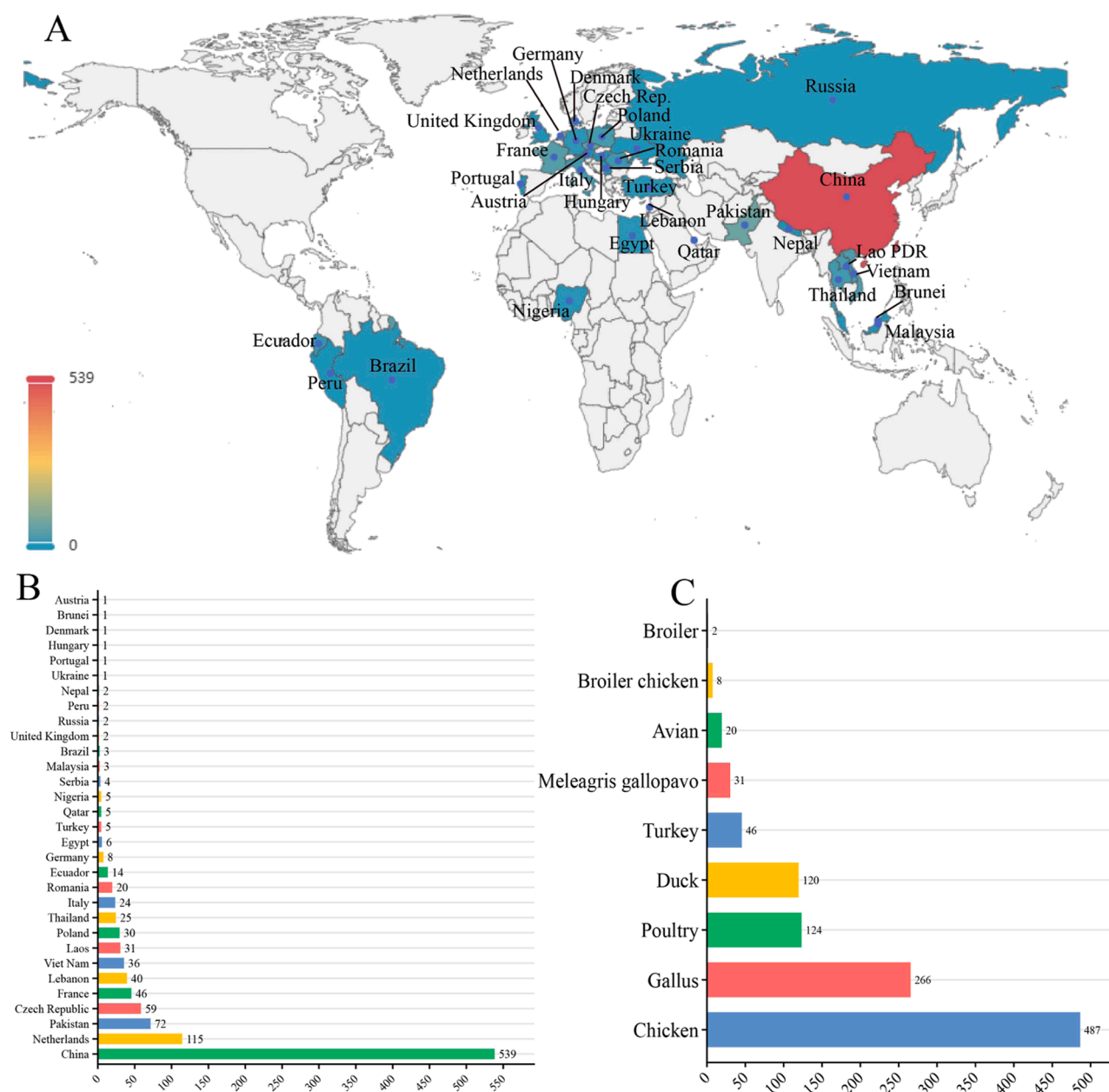


Fig. 1. Sources of avian-derived MCRPEC. (A) Global distribution heatmap of avian-derived MCRPEC. (B) Country-wise distribution and quantity of avian-derived MCRPEC. (C) Host origins and quantities of avian-derived MCRPEC.

strains isolated from Chicken accounting for 44.11 % (487/1104), being the main source of isolation; followed by gallus, accounting for 24.09 % (266/1104). Ducks and poultry have a comparable proportion, at 10.87 % (120/1104) and 11.23 % (124/1104), respectively. The remaining sources account for 9.96 % (107/1104). Looking at the global geographical distribution, the collected MCRPEC come from 31 countries across four continents (Fig. 1A, B), with the highest detection rate in Asia (68.75 %, 759/1104), where China (48.82 %, 539/1104) has the highest detection rate; Europe (28.53 %, 315/1104) has the second highest detection rate, with the Netherlands (10.41 %, 115/1104) being the second highest in detection rate, and the distribution of MCRPEC in Europe is also the widest, involving 13 countries, nearly half of the total number of countries. The remaining isolates come from three countries in the Americas (1.72 %, 19/1104) and two countries in Africa (1 %, 11/1104).

3.2. Genomic characteristics of avian-derived MCRPEC

Based on ClermonTyping, the collected avian-derived MCRPEC strains encompassed eight major phylogroups, with only three strains having unknown types. Group A (44.47 %, 491/1104) and B1 (28.53 %, 315/1104) accounted for 73.01 % of the total. Regarding serotype distribution, the leading serotypes were O153:H34 (2.72 %, 30/1104) and O154:H20 (2.72 %, 30/1104), with no dominant circulating serotypes (Table S1).

Among all avian-derived MCRPEC strains, including *mcr-1*, a total of 180 ARGs were detected. The coexistence of *mcr-1* with other high-risk ARGs significantly increases bacterial resistance levels (Fig.S1). The tetracycline resistance gene *tet(A)* (75.72 %, 836/1104) was the second most prevalent after *mcr-1*, followed by the sulfonamide resistance gene *sul2* (67.03 %, 740/1104). Additionally, TEM family β -lactam resistance gene *bla_{TEM-1B}* (64.04 %, 707/1104), florfenicol resistance gene *floR* (58.15 %, 642/1104), and aminoglycoside resistance gene *aph(6)-Id*

(57.79 %, 638/1104) were also present. Notably, the β -lactam resistance gene *bla*_{NDM-5} (9.24 %, 105/1104) and tetracycline resistance gene *tet* (X4) (2.08 %, 23/1104) were found to coexist with *mcr-1* in a small number of cases.

An analysis of ARGs among the top three contributing countries and host types of avian-derived MCRPEC strains using a t-test revealed a significantly higher prevalence of ARGs in Chinese isolates relative to those from the Netherlands and Pakistan, with notable differences between the latter two (Fig. S2A). Host-specific analysis indicated that chicken-derived isolates possessed a significantly greater number of ARGs compared to those from ducks and turkey, with turkey showing the lowest count (Fig. S2B). Intriguingly, a complementary relationship between ARGs and VFs was observed across the countries and hosts, implying a possible trade-off between the presence of ARGs and VFs. In

terms of ARG diversity, Chinese isolates exhibited the highest number of unique ARGs ($n=70$), whereas the Netherlands and Pakistan had comparatively fewer, with counts of $n=18$ and $n=4$, respectively (Fig. S3A). When considering the host perspective, chicken isolates had the most unique ARGs ($n=72$), ducks had a minimal number of unique ARGs ($n=5$), and turkey isolates shared all ARGs with the other hosts (Fig. S3B). To elucidate the distribution patterns of ARGs, a principal component analysis (PCA) using PCoA was conducted, depicted in Fig. 2A and B. This analysis highlighted a wider spread of ARG types among Chinese isolates, with Pakistani isolates showing overlap with both Chinese and Dutch isolates. Additionally, a comparable distribution of ARGs was observed between chicken and duck isolates, while turkey demonstrated the least diversity in ARG profiles.

Avian pathogenic *E. coli* (APEC) poses a serious threat to the

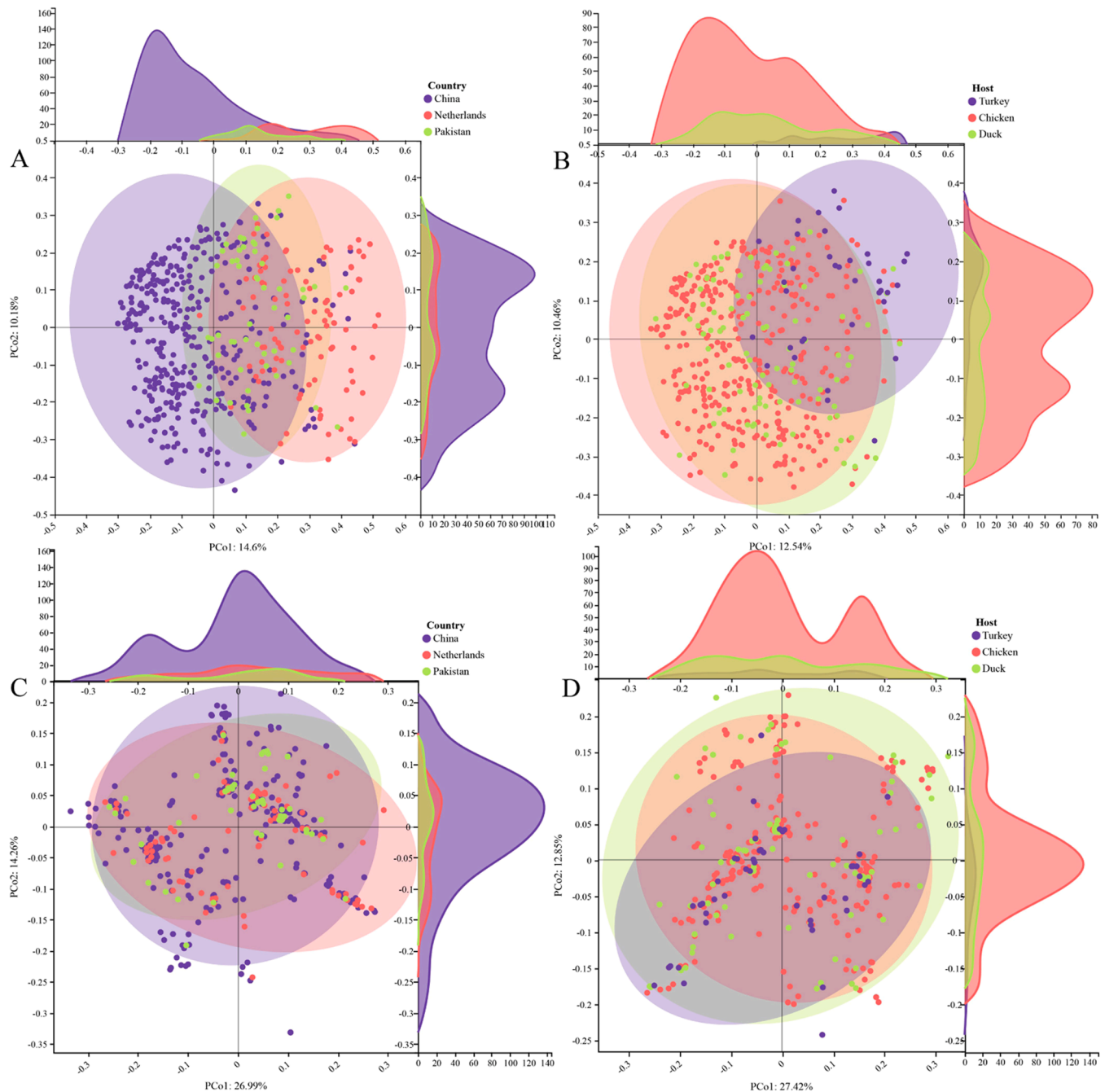


Fig. 2. Principal Coordinate Analysis (PCoA) of ARGs and VFs across different countries and hosts. (A) PCoA of ARGs across three countries. (B) PCoA of ARGs across three host species. (C) PCoA of VFs across three countries. (D) PCoA of VFs across three host species.

development of the poultry industry. The study collected 206 VFs from avian-derived MCRPEC. The fim family genes encoding type I fimbriae were present in 921–1033 strains. P-fimbriae encoding genes related to bacterial colonization were found in 37–210 strains (Table S3). Additionally, genes related to the iron uptake system, *icuA* (n=459) and *iutA* (n=462), invasion factor-related gene *ibeA* (n=32), and the main VFs of Shiga toxin-producing *E. coli*, *stx2A* (n=2) and *stx2B* (n=2), were also identified. From a national perspective, Dutch isolates carried a significantly higher average number of VFs compared to Chinese isolates, with no significant difference observed with Pakistani isolates. Isolates from turkey carried a significantly higher average number of virulence factors compared to those from chickens and ducks (Fig. S2C, D). The principal component analysis of VFs showed similar distributions among the three countries, with China showing slightly higher diversity, and from the host perspective, Duck showed a broader distribution of VFs (Fig. 2C, D). At the same time, MCRPEC also includes pathogenic serotypes such as

O78, O1, and O2, which pose a formidable threat to the health and wellbeing of avian species.

3.3. Diversity of STs of avian-derived MCRPEC

After MLST typing, the 1104 avian-derived MCRPEC strains encompassed a total of 202 sequence types (STs), although 62 strains failed to yield ST typing results. The most prevalent was ST10 (6.34 %, 70/1104), originating from six different hosts across 12 countries, followed by ST156 (4.71 %, 52/1104) and ST48 (4.62 %, 51/1104). In terms of ST diversity, MCRPEC strains isolated from China exhibited the richest variety of STs (56.93 %, 115/202), and among the hosts, strains from chickens displayed the most diverse STs (54.46 %, 110/1104) (Fig. 3A, B). This result indicates that *mcr-1* exists in various STs of *E. coli*, and further research is needed on the differences in *mcr-1* carriage among different STs.

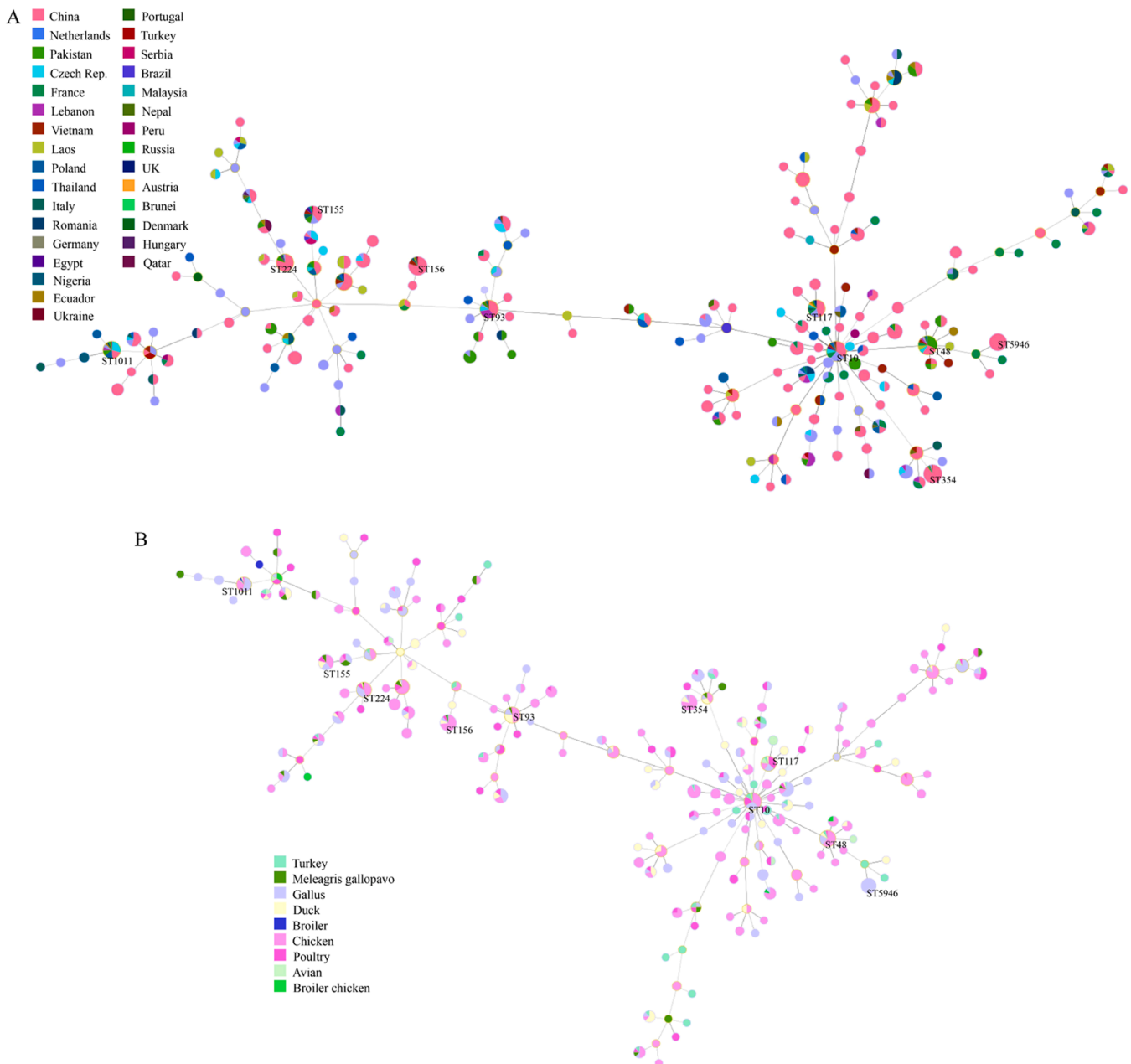


Fig. 3. The distribution of MCRPEC strains by STs originating from poultry sources. (A) Phylogenetic tree based on national-level STs. (B) Phylogenetic tree based on host-level STs.

Selecting the 10 most abundant STs, a comparison of the number of ARGs and VFs they carry revealed that ST156 harbored the highest number of ARGs, while ST10 carried the fewest (Fig. S4A). Similarly, ST10 carried the fewest VFs, whereas ST117 carried the most (Fig. S4B). However, there was no apparent correlation between the number of ARGs and virulence factors across different STs.

To better understand the genetic evolutionary relationships among strains from different sources, pan-genome analysis was conducted on the 1104 avian-derived MCRPEC strains, and the SNP differences between each pair of strains were calculated. Among all strains, a total of 1667 core genes ($99\% \leq \text{strains} \leq 100\%$), 508 soft-core genes ($95\% \leq \text{strains} \leq 99\%$), 4165 accessory genes ($15\% \leq \text{strains} \leq 95\%$), and 73,519 unique genes ($0\% \leq \text{strains} \leq 15\%$) were identified. Phylogenetic tree was constructed based on the SNP differences in the core genome (Fig. 4), where some MCRPEC from different countries and hosts showed evidence of clonal dissemination, such as the German chicken-derived MCRPEC (GCA_001676995.1) with French turkey-derived (GCA_949281455.1), Chinese poultry-derived (GCA_017789485.1), and Chinese avian-derived (GCA_009898445.1) with SNP differences of 74, 46, and 49, respectively (Table S2).

To explore the transmission relationship between avian-derived MCRPEC and environmental and human sources, the same method was used to select 31 *mcr-1*-positive ST10 *E. coli* strains from environmental and human sources from the NCBI database (Table S9). These were combined with 70 MCRPEC strains from this study to construct a phylogenetic tree based on SNP differences in the core genome (Fig. 5A). The figure shows that some avian-derived MCRPEC cluster with human-derived MCRPEC, originating from different countries, indicating a close phylogenetic relationship. Avian-derived MCRPEC generally carry a higher number of ARGs. The only environmental source MCRPEC collected showed a distant phylogenetic relationship with the other strains. Notably, avian-derived GCA900093965.1, human-derived GCA001940485.1, and human-derived GCA021067405.1 may exhibit clonal dissemination ($64 < \text{SNP} < 118$) (Table S4), and all three strains come from different countries. To

determine if there is any correlation between the number of ARGs and VFs, ISs, and plasmid replicons, a Spearman analysis was performed on 70 avian-derived *mcr-1*-positive *E. coli* strains (Fig. 5B, C&D). The number of ARGs was significantly correlated with the number of plasmid replicons and ISs, but not significantly correlated with the number of VFs.

3.4. Pan-GWAS analysis of avian-derived MCRPEC

To understand the differences in accessory genes among the three countries and hosts with the highest number of strain sources, we conducted a pan-GWAS study on strains from China, the Netherlands, and Pakistan, and from chickens, ducks, and turkeys. The most differential genes were found between chickens and Chinese sources of MCRPEC (Fig. S5). Chicken-derived MCRPEC had 835 significantly differential genes, coding hypothetical proteins accounting for 64% (535/835). Apart from genes related to vital life activities, most are encoding genes associated with ISs, transposons, and the type IV secretion system, such as *traG* and *traL* (Table S6). The abundance of these encoding genes may have accelerated the dissemination of ARGs. The type IV secretion system *virB1*, as a marker differential gene, appears in chicken-derived MCRPEC. Its role in the colonization and growth of MCRPEC in chickens or in the spread of *mcr-1* requires further research. MCRPEC originating from China, apart from hypothetical proteins, had 463 differential genes, and more differential genes related to multidrug resistance efflux pumps appeared compared to Chicken sources, suggesting a reason for the general difference in the drug resistance spectrum of Chinese-derived MCRPEC (Table S5). Duck and turkey sources had fewer differential genes, which are basically genes related to life activities, possibly the result of *E. coli*'s adaptive evolution within the host. The marker differential gene of Pakistani-derived MCRPEC is *flmC*, which is closely related to the flagella synthesis of *E. coli* in that region, suggesting a stronger motility. In contrast, Dutch-derived MCRPEC had more differential genes, but most were hypothetical proteins, among which the bicyclomycin resistance gene showed a significant difference

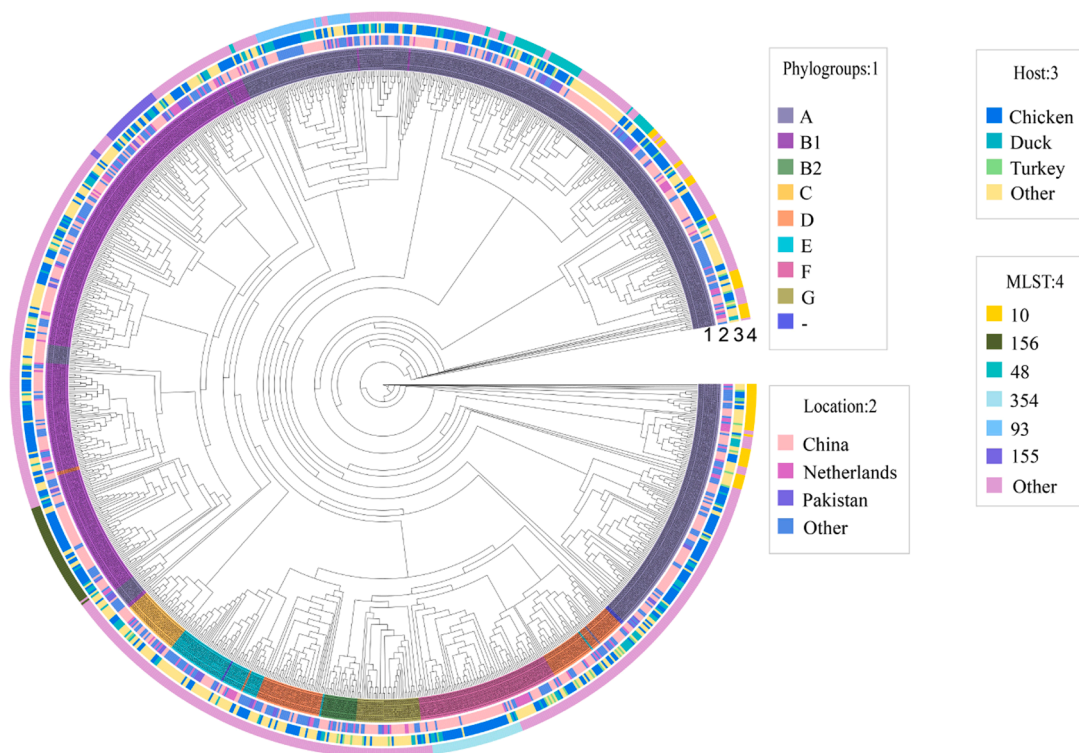


Fig. 4. Phylogenetic tree based on core gene SNP of all strains, with Circle 1 indicating the Clade type, Circle 2 denoting the country of origin, Circle 3 representing the host source, and Circle 4 indicating the Sequence Type analysis.

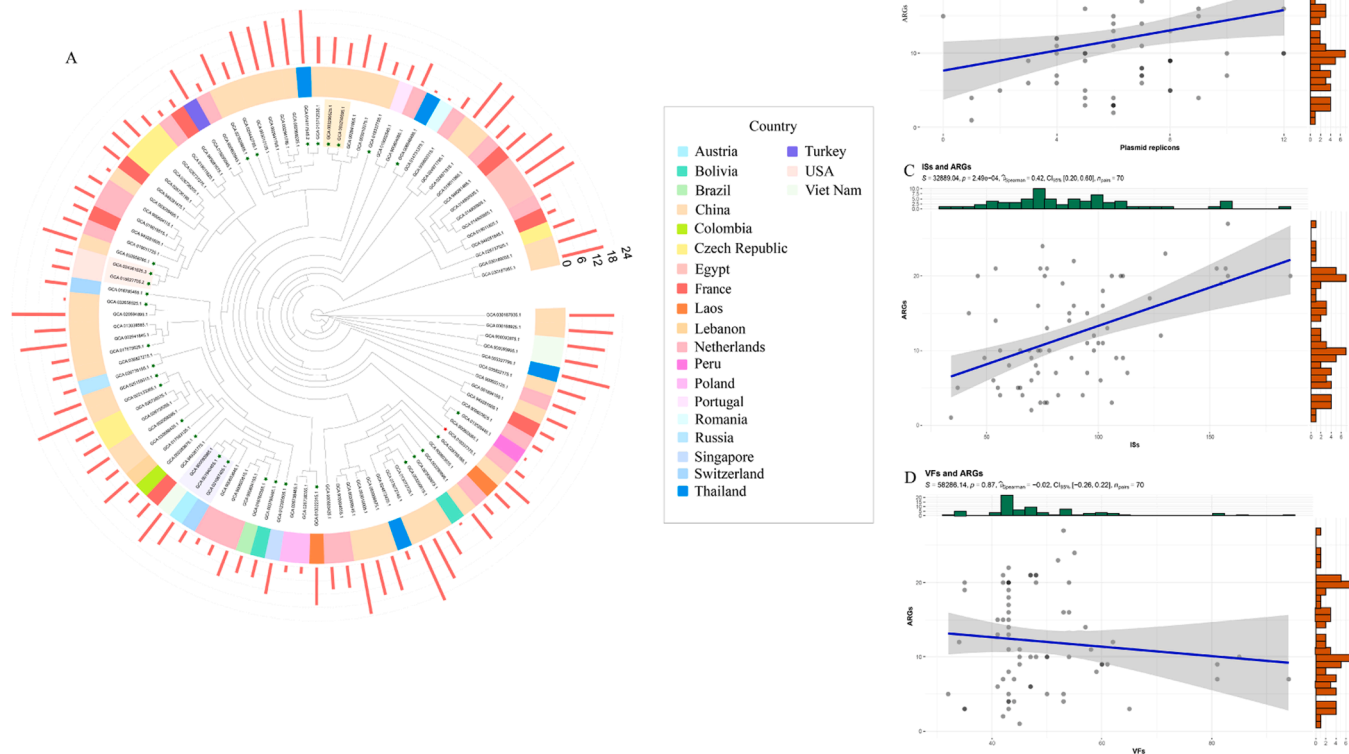


Fig. 5. Analysis of ST10 avian-derived MCRPEC. (A) Phylogenetic tree of ST10 avian-derived MCRPEC based on core genome SNPs, the outermost ring bars represent the number of ARGs carried. (B) Spearman correlation analysis between ARGs and plasmid replicons. (C) Spearman correlation analysis between ARGs and ISs. (D) Spearman correlation analysis between ARGs and VFs.

from *E. coli* from other countries. To understand the situation of differential genes carried by MCRPEC carrying *tet(X4)* and *bla_{NDM-5}*, pan-GWAS analysis was performed respectively. Compared with *mcr-1*-positive bacteria carrying *bla_{NDM-5}*, those carrying *tet(X4)* had more type IV secretion system encoding genes (*traI*, *traS*), and the common upstream and downstream insertion sequences of *bla_{NDM-5}* appeared as differential genes (Table S7).

3.5. Genetic environment analysis of *mcr-1*

Based on the analysis results from Plasclass, we conducted a genetic environment analysis of the *mcr-1* located on plasmids (Fig. 6A). A total of 262 avian-derived MCRPEC strains had the *mcr-1* gene located on plasmids ($p > 0.8$) (Table S8). The flanking genes of the plasmid-located strains were trimmed, and redundancy was removed using cdhit, resulting in 12 different *mcr-1* flanking environments. The most common flanking environment was IS*ApI1*, including the most classic Tn6330, as well as truncated Tn6330, where the absence of IS*ApI1* at the upstream and downstream positions, forming a single IS*ApI1* cap, is also common. Non-classic Tn6330 structures (GCA_900603125.1) showed truncation of IS*ApI1*. At the same time, we observed a rich variety of *mcr-1* flanking environments, the role of which in the transfer of *mcr-1* requires further research.

Combining the prediction results from Plasclass, we utilized mob-suite for plasmid typing. The reconstructed plasmids carrying *mcr-1* were identified for type and mobility behavior prediction (Table S8). Among them, 249 plasmids carrying *mcr-1* were predicted to be transferable through conjugation. The prevalence of plasmids carrying *mcr-1* was analyzed based on host and country of origin (Fig. 6C), with

chickens primarily using IncI2 as the main vectors for *mcr-1*, and similarly, China and the Netherlands are prevalent in IncI2 and IncX4. From the perspective of the popular plasmid types carrying *mcr-1*, IncI2 is the main popular plasmid, followed by IncX4. In addition, IncHI2A, IncHI1B, IncFIB, and IncFIA are also sporadically spreading. We divided the stages into four phases according to the collection date of the sequences and found that in the early stage (2004–2009), only a small number of plasmids carrying *mcr-1* appeared, and the types of plasmids were relatively singular, mainly IncI2. In the middle stage (2010–2014), the types of plasmids became diverse and the number carrying *mcr-1* increased (Fig. 6B). In the later stage, the plasmids carrying *mcr-1* became even more diverse, but compared with the middle stage, some plasmids carrying *mcr-1* no longer appeared. IncI2 has been carrying *mcr-1* since it first appeared and has been steadily present in every period, with its quantity continuously increasing, becoming the main prevalent plasmid carrying *mcr-1*. IncX4 also appeared in the middle stage, and the number of positive plasmids was second only to IncI2.

4. Discussion

Compared to previous studies, our collection of MCRPEC is more extensive, encompassing nearly all high-quality genomic data for avian sources. In contrast to review articles that summarize regional studies and derive patterns, our concrete analysis of genomic data better elucidates the prevalence and genomic characteristics of MCRPEC in avian sources. Through this study, we have gained a preliminary understanding of the geographical distribution and genomic characteristics of avian-derived MCRPEC. A series of analyses were conducted focusing on genomic features, and the results confirmed that avian-derived MCRPEC

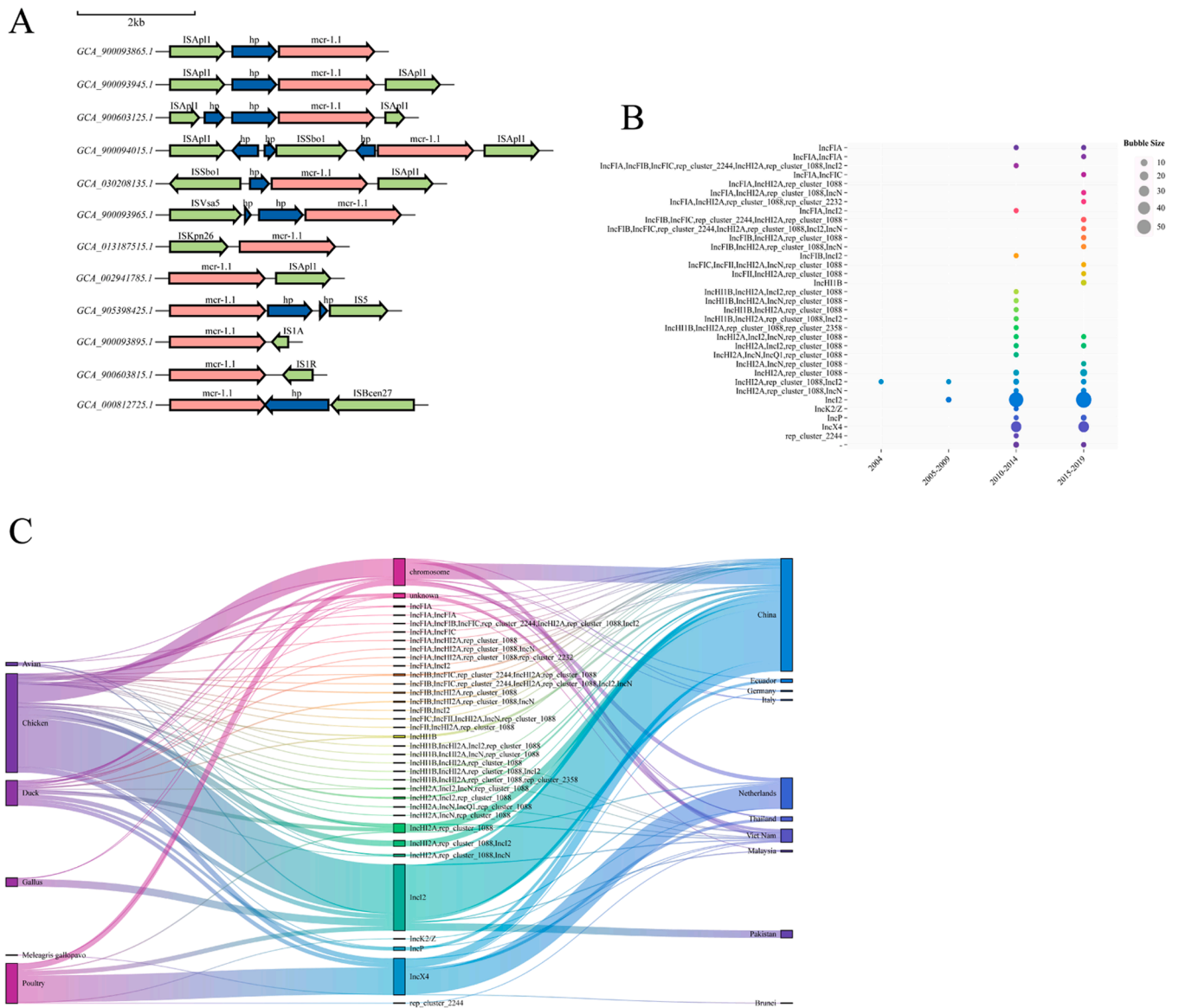


Fig. 6. The flanking environment and the prevalence of positive plasmids of *mcr-1*-positive *E. coli*. (A) Linear alignment map of the avian *mcr-1* flanking environments. (B) The prevalence of *mcr-1*-positive plasmids across four periods. (C) The distribution of *mcr-1*-positive plasmids in different countries and hosts.

are widely distributed globally, involving most European countries, which may be related to the frequent poultry trade among these nations. China has the highest number of positive strains, associated with the extensive use of colistin prior to 2017, a significant factor in promoting the proliferation and spread of MCRPEC. Nearly half of the countries have no more than 5 strains of MCRPEC, which is related to the regulation of polymyxin use in the poultry industry (Webb et al., 2017). In addition, this is related to the different levels of attention paid to MCRPEC. The larger the amount of data in a country, the better the analysis can reflect the actual situation. Most ARGs can co-occur with *mcr-1* in the same strain, with the most common being *tet(A)* and the *sul* family of ARGs. Among these are also high-concern ARGs such as *bla_{NDM}* and *tet(X)*, which pose a significant threat. Research has shown that this type of *E. coli* has been found in both human clinics and animals, carrying ARGs that not only endow bacteria with strong resistance, but also pose a risk of transmission (Mediavilla et al., 2016; Lu et al., 2022). In terms of ARGs abundance, Chinese isolates have the widest abundance and the most unique genes. Due to the fact that some of the collected data in this study were only classified as birds without specific host details, the relevant analysis was based on chickens, ducks, and turkeys

with clear host types. In terms of ARGs quantity, turkeys are significantly lagging behind chickens and ducks, which may be related to their different reproductive patterns compared to regular poultry. The number of *mcr-1*-positive strains from chickens is the highest and has the most unique ARGs, with a similar abundance of ARGs as duck-derived *mcr-1*-positive strains. In addition, we found that there is a certain degree of complementarity between the number of ARGs and VFs in the relationship between countries and hosts. In this study, it was observed that MCRPEC derived from poultry typically carry multiple VFs, and common pathogenic serotypes of APEC, such as O78, O1, and O2, also exhibit *mcr-1* positivity (Hu et al., 2022).

From the SNP analysis based on the core genome, it can be observed that there has been a large-scale clonal dissemination phenomenon between isolates of avian-derived MCRPEC collected in different years within China. Some strains from China, the Republic of Zambia, Nigeria, Romania, Lebanon, Italy, Poland, the Netherlands, France, and other countries have SNPs less than 200, indicating a global clonal dissemination phenomenon. We hypothesize that the transmission of avian-derived MCRPEC across hosts and regions may be influenced by factors such as animal migration, human activities, and trade transactions,

with these dissemination events often exhibiting correlated patterns. The use of colistin is also likely to exacerbate the spread. Prior research has suggested that human-derived MCRPEC in China is predominantly linked to the transmission through food sources, while the potential for waterborne transmission of MCRPEC warrants consideration (Shen et al., 2018). These factors may play a role in facilitating the extensive spread of avian-derived MCRPEC within China. To clarify the global transmission pathways of avian-derived MCRPEC, ongoing surveillance of breeding environments, production and transportation processes, and trade practices is essential to identify the primary routes of dissemination. Avian-derived MCRPEC possess a rich diversity of STs, with ST156 carrying a significantly higher number of ARGs, playing an important role in the spread of ARGs. ST117 carries a significantly higher number of VFs, potentially posing a pathogenic risk to the host. The most widely disseminated is the multi-clonal ST10 type, which is mostly clustered with human-derived MCRPEC in phylogenetic relationships, showing a close kinship, with some strains having SNPs less than 200. A Spearman analysis was conducted on the avian-derived ST10 MCRPEC, and the results indicated a positive correlation between the number of ARGs and the number of ISs and plasmid replicons, with no correlation with the number of VFs. This result is consistent with the analysis of ST10 type *tet* (X)-positive *E. coli* (Li et al., 2023).

Based on pan-GWAS, analyzing the differential accessory genes between strains from different sources can help us understand the unique accessory genes carried by a particular source strain during the process of adaptive evolution. Here, we selected single characteristics such as different countries and host sources for pan-GWAS analysis with the population. The conjugative transfer proteins TraG and TraL, as differential genes, appear in chicken-derived MCRPEC, the type IV secretion system is currently the only known system capable of mediating the horizontal transfer of ARGs and plays an important role in the dissemination of ARGs (Ding et al., 2022). Duck-derived MCRPEC also exhibit differences in the Type IV secretion system protein, with traL appearing as a differential gene, which may explain why the number of ARGs in chicken-derived is significantly higher than in duck-derived and turkey-derived, and duck-derived are significantly higher than turkey-derived. China-derived MCRPEC have a rich diversity of differential gene functions, mainly related to efflux pump genes, which may be related to the high-pressure antibiotic environment faced by China-derived MCRPEC (Qiao et al., 2018), allowing *E. coli* to retain and stably inherit the encoding genes related to efflux pumps during adaptive evolution. At the same time, it was also observed that *bla*_{NDM-1} appears as a signature gene in China-derived MCRPEC, which is consistent with the aforementioned *bla*_{NDM-1} being a unique ARG in China-derived MCRPEC, and the commonly MGEs upstream and downstream of *bla*_{NDM-1} also appear as characteristic genes (Osei Sekyere et al., 2024). Compared to MCRPEC from China, MCRPEC from Pakistan's avian sources also possess IS903 and IS_{Psy42} as distinct MGEs, which may play a role in the dissemination of certain ARGs. However, the differential genes of Netherlands-avian MCRPEC, apart from being related to vital activities, mainly involve genes related to pilin proteins. At the same time, the signature differential genes of Pakistan-avian MCRPEC are related to flagella, which may increase the motility and adhesion ability of the MCRPEC from this source, which is conducive to its own spread and is related to virulence. In addition, strains carrying *tet* (X4) also possess two types of IV secretion system proteins, traL and traS, which may facilitate the spread of *tet* (X4) and some ARGs, and further exploration is needed.

In order to understand the genetic environment of *mcr-1* in avian derived MCRPEC, we trimmed and clustered the flanking regions of *mcr-1* located on the plasmid. We observed 12 flank environments, most of which are Tn6330 structures. Current research has demonstrated that the *mcr-1* gene, positioned within the Tn6330 transposon, constitutes a stable and potent mechanism for genetic mobilization. This study has identified other defective Tn6330 structures. However, when the *mcr-1* gene is missing the downstream IS_{Apl1} element, it is incapable of being

mobilized. (Snesrud et al., 2018). IS_{Kpn26} is also often present upstream and downstream of *mcr-1* and is considered to be related to *mcr-1* resistance, with *Klebsiella pneumoniae* carrying IS_{Kpn26} being more challenging to treat (Yang et al., 2020). Extant research has unearthed the presence of IS_{1R} in the genomic vicinities of foodborne *Cronobacter* isolates and human clinical *E. coli* strains (Zhu et al., 2023). Notably, within these *E. coli* isolates, an IS_{1A}-mediated VF exists, distinguishing it from the context of IS_{1A}'s involvement in the *mcr-1* flanking environment reported in this study. The ISs identified in our study (IS_{Vsa5}, IS_{Apl1}) demonstrate a higher propensity for insertion into the *mcr-1* gene architecture, ultimately leading to its silencing, rather than serving as mobile elements facilitating *mcr-1* dissemination (Fan et al., 2020). Furthermore, IS₅, IS_{1A}, IS_{1R}, and IS_{Apl1}, among others, have frequently been reported to insert into the *mcrB* gene, resulting in heightened polymyxin tolerance, underscoring the intricate interplay between these ISs and polymyxin resistance mechanisms (Shamina et al., 2020; Azam et al., 2021). Upon reclassification of the plasmids, a substantial proportion of *mcr-1*-harboring plasmids (n=249) were predicted to be conjugatively transferable, posing a formidable risk to public health. Our investigation revealed disparities in the prevalence of *mcr-1*-positive plasmids among avian *E. coli* isolates sourced from diverse countries and hosts. Specifically, IncI2 plasmids exhibited greater dominance in China and chicken-origin isolates, whereas IncX4 plasmids showed a distinct prevalence advantage in the Netherlands, mirroring findings from localized surveys conducted within this region (Jamin et al., 2021). These findings have significant implications for the development of targeted intervention strategies in veterinary medicine and public health. IncI2, a predominant vector for the *mcr-1*, utilizes the *pcnR* (plasmid copy number repressor) to control plasmid replication, thereby fine-tuning the expression of *mcr-1* and bolstering the bacterium's adaptive capacity (Yang et al., 2021). This regulatory mechanism has positioned IncI2 as one of the most frequently identified plasmids for the carriage and transfer of *mcr-1*. Similarly, the IncX4 plasmid contains the conjugative transfer stimulon *pixR*, which activates the expression of a critical set of transfer genes by binding to their promoters. Research indicates that *pixR* is crucial for the enduring presence of *mcr-1*-positive IncX4 plasmids and enhances their conjugative transfer efficiency, making IncX4 a favored vector for *mcr-1* in natural environments (Yi et al., 2022). Upon reviewing the timeline, it is evident that prior to 2009, there were only a few *mcr-1*-positive plasmids detected, and these belonged to a limited number of plasmid types. However, starting in 2010, there was a notable increase in the number of *mcr-1*-positive plasmids, accompanied by the appearance of multiple plasmids carrying this gene. By 2019, avian-derived MCRPEC continued to harbor a large number and a diverse range of *mcr-1*-positive plasmids. When comparing these changes to the period following China's discontinuation of colistin usage, it is observed that globally, there has not been a significant decrease in the number of *mcr-1*-positive plasmids in avian-derived MCRPEC. However, some specific plasmid types have disappeared, while new *mcr-1*-positive plasmids have emerged (Shen et al., 2020). Notably, since the emergence of the IncI2 plasmid carrying *mcr-1*, its prevalence has steadily risen, becoming the predominant plasmid type and playing a crucial role in the dissemination of *mcr-1* (Li et al., 2020). This underscores the ongoing challenge of antimicrobial resistance and the importance of continuous monitoring and strategic interventions in veterinary medicine.

5. Conclusion

This study conducted a global analysis of avian-derived MCRPEC from the NCBI database, focusing on the genomic data to support monitoring and controlling the spread of *mcr-1* and its public health implications. Chinese and chicken-derived MCRPEC strains had the highest number of *mcr-1* positives and carried a diverse array of ARGs. There was evidence of widespread clonal dissemination of MCRPEC across countries. The strains exhibited a rich diversity in ST types, with

ST10 being predominant, and showed positive correlations between ARGs, ISs, and plasmid replicons. Pan-GWAS analysis revealed that differential genes in MCRPEC are mainly associated with type IV secretion systems, ISs, and transposons. The genetic environment of *mcr-1* is predominantly within Tn6330, with various ISs in its flanking regions potentially promoting its spread. The prevalence of *mcr-1*-positive plasmids suggests a significant link between IncI2, IncX4, and *mcr-1*, which could help its transmission. The study acknowledges limitations in data representativeness and the impact of uneven data sources. It concludes that *mcr-1* is widely distributed in avian sources across many countries, posing a significant risk and threat, necessitating long-term monitoring and strict control measures to safeguard public health and support the poultry industry.

Author contributions

QL and SZ wrote this manuscript; JY, QL, YH, ZW and SZ contributed to the design of this manuscript and perform the experiments; MW, RJ, ML, DZ, SC, and YW provided ideas for the conception of this manuscript; XO, QY, BT, DS, XZ, and JH helped to collect samples; AC and SZ modified this manuscript, and acquired Funding; All the authors reviewed the manuscript.

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CRediT authorship contribution statement

Anchun Cheng: Writing – review & editing, Methodology, Investigation, Funding acquisition. **Mafeng Liu:** Methodology, Investigation. **Zhen Wu:** Methodology, Investigation. **Shun Chen:** Methodology, Investigation. **Xinxin Zhao:** Visualization, Data curation. **Shaiqu Zhang:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Dekang Zhu:** Visualization, Data curation. **Ying Wu:** Visualization, Data curation. **Juan Huang:** Visualization, Data curation. **Qiao Yang:** Visualization, Data curation. **Di Sun:** Visualization, Data curation. **Jing Yang:** Methodology, Investigation, Data curation. **Xumin Ou:** Visualization, Data curation. **Qianlong Li:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Yu He:** Investigation, Formal analysis. **Renyong Jia:** Methodology, Investigation. **Bin Tian:** Writing – original draft, Data curation. **Mingshu Wang:** Writing – review & editing, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. None

Data Availability

Data will be made available on request.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ecoenv.2024.117109](https://doi.org/10.1016/j.ecoenv.2024.117109).

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